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Infrastructure Asset Management

Artificial intelligence in healthcare facilities asset information management: mixed review

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Healthcare facilities are pivotal in ensuring continuous access to services, particularly for individuals with complex health conditions. Effective asset information management (AIM) in these facilities through artificial intelligence (AI) can enhance operational efficiency. The exploratory bibliometric and systematic review assesses the status and trends of AI applications in AIM within healthcare facilities. The findings reveal a significant gap between research findings and practical implementation, highlighting the need for further integration and real-world usage of AI-powered solutions in healthcare facilities settings. This study identified notable gaps, including the need for research on utilising AI to enhance asset management in healthcare, including maintenance scheduling and procurement processes. Involving stakeholders, such as healthcare professionals, facility managers, and patients, in effective facility management using AI requires further investigation. Research is needed to evaluate the economic benefits and develop robust ethical guidelines for responsible AI implementation. Notably, previous research has given limited attention to AI for healthcare AIM, with emerging trends focusing more on AI and infrastructure than the 'asset' aspect. Implementing AI-powered solutions tailored to the unique needs of healthcare facilities and evaluating cost-effectiveness will lead to improved asset management practices, enhanced decision-making processes, and, ultimately, more efficient and effective healthcare operations.

Keywords: artificial intelligence/asset information management/exploratory review/healthcare facilities/UN SDG 9: industry, innovation and infrastructure/UN SDG 11: sustainable cities and communities

Introduction

Healthcare facilities management is crucial to providing healthcare services. These facilities include hospitals, homecare services, healthcare centres, and clinics (Li *et al*., 2021). Mahadkar *et al*. (2012) noted that managing the physical assets within healthcare facilities is essential for delivering healthcare services and improving productivity. In addition, effective healthcare asset management ensures the functionality of services by considering various processes such as installation, maintenance, and risk management (Vilcahuamán and Rivas, 2017). However, healthcare facilities worldwide face rapid deterioration, a global phenomenon underscored by Ahmed *et al*. (2020). Factors such as insufficient finance, compromised quality control, and inadequate maintenance contribute to this decline, causing these facilities to deteriorate more rapidly than they can be replaced.

When a component can no longer be depended upon to perform its primary duties, inadequate maintenance across the asset life cycle increases the likelihood of asset breakdown and accelerates deterioration (Heng and Loosemore, 2013). A critical aspect of facilities management is prioritising maintenance and rehabilitation interventions, considering urgency, predicted improvements, costs, and downtime, especially for healthcare facilities that operate continuously. Therefore, reliable and up-to-date assessments of the state of healthcare facilities are essential for effective facilities management (De Jager and Wall, 2022).

Effective healthcare asset management requires dual data quality and data coverage. The data quality ensures accuracy and reliability in decision making through comprehensive asset management strategies (Adoma *et al*., 2023; Ravikumar *et al*., 2022). According to Abdirad and Dossick (2020), the major issues for operators and owners, such as the government and private sectors, are creating facility asset data and record models. These challenges include collecting accurate data about facility assets, integrating data from disparate systems, ensuring data accuracy, and standardising the data. All these processes can be complex and time-consuming, particularly for organisations with extensive and diverse asset

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portfolios. Organisations that cut through proper processes and procedures lead to catastrophic events (Deepa *et al*., 2020).

The absence of accurate asset data and record models made it difficult to identify critical maintenance needs, leading to the tragic collapse (Salem and Helmy, 2014). More recently, a fire broke out on 16 April 2021 at Charlotte Maxeke Johannesburg Academic Hospital in Johannesburg, South Africa. The fire started in the early morning and spread quickly, destroying parts of the hospital's main building (Tshalibe *et al*., 2023). The fire at Charlotte Maxeke Johannesburg Academic Hospital significantly impacted the hospital's infrastructure. The inadequate data on asset acquisition, maintenance, performance, and disposal made it challenging to assess the full extent of the damage. The limited data significantly hindered the assessment of the damage's extent, tracking and ordering of replacement parts, and coordinating with insurance companies for claims and reimbursements, creating a challenging recovery process. Such deficits in data regarding the life cycle of assets, their usage, repair histories, and operational efficacies can impede effective management, predictive maintenance, and strategic planning, ultimately impacting the quality and continuity of healthcare services.

Traditional maintenance practices must be systematically modified, while traditional design and construction practices must also be considered to enhance performance concerning facilities and the environment, especially in developing countries (Rani *et al*., 2015). Regular inspections, preventive maintenance, and timely repairs improve the infrastructure's operational efficiency and functionality, extending the facility's lifespan and mitigating costly interventions (Mangano and de Marco, 2014). Extending the lifespan of facilities reduces the need for costly replacements and renovations. These modifications result in cost savings, benefiting the overall development and well-being of the facility. Traditional maintenance practices often face challenges in effectively managing facilities and the environment, particularly in developing countries. As a result, the performance of facilities and infrastructure can suffer, leading to increased downtime, decreased productivity, and negative environmental impacts. To overcome these challenges and improve maintenance practices, integrating technology, such as artificial intelligence (AI), can play a crucial role in addressing challenges.

AI has emerged as a potential option for creating, combining, and maintaining integrated databases, which house the pertinent data for a facility (to support operations and maintenance (Cavka *et al*., 2017)). AI has been widely described as the development of intelligent agents that enable computers to think, act, and gather analytical data to enhance organisational management. In the Architecture, Engineering, Construction, and Operations sector, AI has the potential to facilitate electronic communication, display sensor data, swiftly simulate scenarios, accurately predict the consequences of hypothetical situations, improve outcome anticipation, and provide more effective instructions for governing the physical environment (Farghaly and Hagras, 2022). While AI has the potential to streamline and enhance various aspects of the facilities management workflow, from defining information requirements to creating the asset information management (AIM), its successful implementation hinges on addressing several key challenges, including data quality, system integration, customisation, security, human collaboration, financial investment, trust, and ethical considerations.

Previous research has examined the application of AI in healthcare asset management. This encompasses the utilisation of building information modelling (BIM) to enhance built asset management (BAM) (Wanigarathna *et al*., 2019). BIM is not just software or a 3D model; it is a process that involves the generation, management, and collaboration around the digital representation of physical and functional characteristics of places. This digital representation serves as a shared knowledge base for information about the facility, which can be used throughout its life cycle (Eastman *et al*., 2011). Other studies have looked at improving fire safety management (Agus Salim *et al*., 2021), optimising performance management (Amos, 2022), and implementing reliability-centred maintenance (Shamayleh *et al*., 2020). The authors have found a gap in research regarding the use of AI as a support tool for healthcare asset management. They highlight a lack of comprehensive articles that thoroughly investigate this application of AI in healthcare.

There is a need for a comprehensive overview of the current state of research in this field. This indicates the need for further studies and evidence-based insights to understand the potential benefits and challenges of utilising AI for asset management in healthcare. This involves the responsible deployment of AI to benefit end users while maintaining accountability and explaining the underlying factors behind AI-driven decisions. The carefully collected and analysed data and results will help healthcare facility operators and owners understand how AI can improve hospital infrastructure's efficiency, effectiveness, and sustainability.

The exploratory bibliometric review on leveraging AI for effective healthcare asset information management serves multiple purposes. It not only contributes by identifying trends, evaluating research output, and visualising the research landscape but also provides valuable insights into the current state of research. By analysing the literature, this review highlights research gaps. It identifies key contributors in the field, allowing researchers and practitioners to understand the relationships between different concepts and themes within the domain. The objectives of the study are as follows.

- Determine the status quo of AI for effective healthcare asset information management.
- Identify trends and themes prevalent in AI for healthcare AIM.
- Explore opportunities and challenges associated with implementing AI in AIM within healthcare.

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- Identify research gaps that require further investigation and advancement in AI applications for healthcare facilities' asset information management.

Section 1 provides the necessary context for the research, outlining AI in healthcare facilities' AIM. Section 2 details the bibliometric and systematic methods to collect and analyse the relevant articles. Section 3 presents the findings of the study. Section 4 delves into a comprehensive discussion of the key points identified in the findings. Section 5 concludes the study by summarising the main findings and providing a reflective overview of the research.

Asset information management

AIM is a strategic process of collecting, organising, storing, controlling, analysing, securing, sharing, and reporting asset information within a virtual system for effective management-based decisions. In facilities management, AIM involves managing the information related to built assets throughout their life cycle using AI technologies and asset management systems (Munir *et al*., 2020). AIM is proactively managing infrastructure assets through various processes or activities. It entails keeping track of information on asset purchase prices, original and remaining service lives, physical condition, and the history of repair and maintenance expenditures. It also involves planning for asset upkeep, repair, and replacement (Cecconi *et al*., 2017). Salem and Elwakil (2020) noted that failure due to poor asset management could result in fatalities, serious injuries, and financial losses. The performance of infrastructure assets must be continuously maintained. Healthcare AIM refers to managing and maintaining information related to the assets used in healthcare facilities and their built infrastructure. This includes tracking each asset's location, condition, and maintenance history and managing the procurement and disposal of assets. Effective asset information management can help healthcare facilities optimise their asset utilisation and reduce costs (Castro *et al*., 2013).

Asset information management and artificial intelligence

AIM in healthcare facilities refers to collecting, organising, and utilising data related to various assets within the facility, such as medical equipment, infrastructure, and resources. It systematically manages asset information to ensure optimal utilisation, maintenance, and planning. On the other hand, AI in healthcare facilities refers to applying intelligent systems and algorithms to automate tasks, analyse data, and make informed decisions. AI technologies can enhance asset information management by automating data collection, enabling advanced analytics and predictive modelling, and facilitating intelligent asset tracking and inventory management (Gbadamosi *et al*., 2021). AIM and AI are crucial in improving operational efficiency, optimising resource allocation, and enhancing decision making in healthcare facilities (Wang and Wu, 2021).

Advanced data analytics and predictive modelling are also made possible by AI. AI systems may analyse previous data to find trends, forecast asset failure or maintenance requirements, and optimise resource allocation using machine learning (ML) techniques (Mattioli *et al*., 2020). Two primary ML techniques are used in this context: supervised and unsupervised learning. Supervised learning involves training a model on labelled data, where the input–output pairs are known (Sen *et al*., 2020). This type of learning is commonly used in predictive maintenance and asset tracking. Unsupervised learning deals with unlabelled data, where the model tries to find hidden patterns or intrinsic structures (Usama *et al*., 2019). This approach is useful for clustering and anomaly detection without predefined labels. Supervised and unsupervised models help allocate resources efficiently based on asset performance and utilisation patterns.

According to Aransyah *et al*. (2020), the lifespan of assets may be extended while downtime is minimised thanks to these features. AI helps optimise asset utilisation and detect inefficiency, saving costs and increasing operational effectiveness. AI can improve decision making in asset management by offering insightful analysis and counsel (Kuziemski and Misuraca, 2020). This allows healthcare facility managers to decide intelligently on asset upkeep, replacement, and investment. These AI technologies automate asset identification and classification within healthcare facilities, expedite inventory procedures, and enhance supply chain management.

Materials and methods

This paper aims to investigate the use of AI as a tool for healthcare information management, specifically in the context of hospital infrastructure and AIM. A systematic literature review (SLR) approach and bibliometric literature analysis have been used to map the best practices in scientific knowledge (Tjebane *et al*., 2023). Together, these two complementary approaches use quantitative bibliometric tools to depict the development of scientific understanding in a particular field and systematic qualitative reviews to conduct in-depth analyses of topics and contents (Pizzi *et al*., 2020). Figure 1 shows the research approach used in this article.

Systematic literature review

An SLR systematically collects available evidence and evaluates the evidence against predetermined criteria (Onososen and Musonda, 2022). The first phase consists of three main steps: defining the scope of the analysis to establish the study's boundaries, finding studies based on set criteria, and selecting and evaluating studies to determine the most pertinent ones (Comerio and Strozzi, 2019).

Scopus was used as the database for extracting the publications. Scopus was selected over other databases as it has engineering, environmental, and computer science publications that others do not (Cavacini, 2015). Scopus has a broader coverage of publications on

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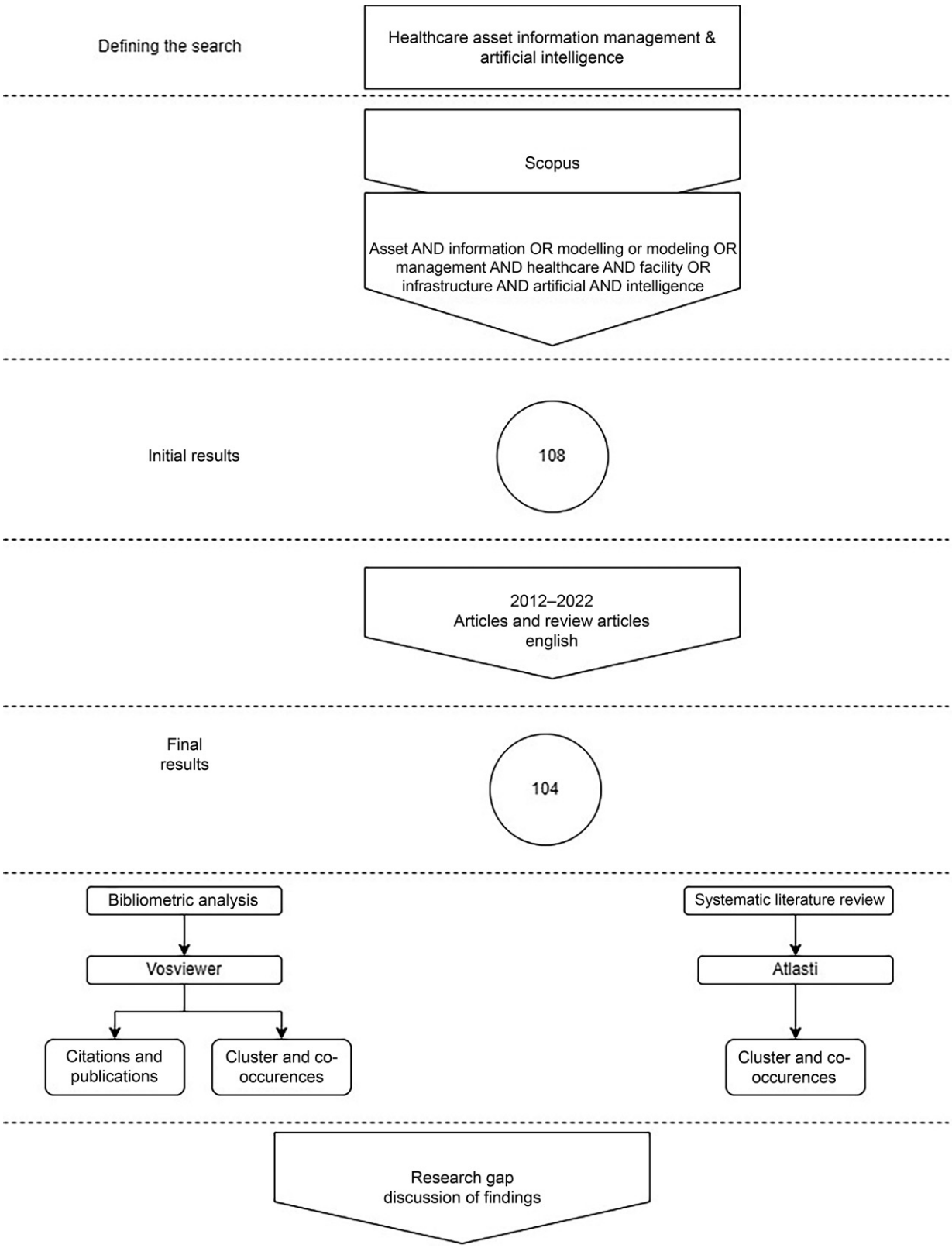

Figure 1. Flowchart of the research method process

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facilities management and built environment project management than Web of Science, EBSCOhost, or ProQuest. It also includes recent publications (Aghaei Chadegani *et al*., 2013). Scopus's citation metrics are widely used and accepted within the research community (Levine-Clark and Gil, 2021).

This research theme used a combination of keywords to find relevant studies (Bartolini *et al*., 2019). The keywords used are 'asset AND information or modelling OR modelling OR management AND healthcare AND facility OR infrastructure AND artificial AND intelligence'. The keywords were searched through Boolean operators 'AND' and 'OR' (Manzoor *et al*., 2021). Boolean operators are words used in conjunction with keywords in a search to combine or exclude them, producing more targeted and valuable results. They aid in removing irrelevant hits that need to be scanned before being discarded. The operators greatly reduce or expand the number of records returned (Nneka Eke-Okpala and Omekwu, 2014).

For this article, we looked at publications made between 2012 and 2022, a decade. According to Alotaibi *et al*. (2020), AI has been extensively studied in the literature due to the rapid advancements in automation, especially over the past 10 years. Moreover, over the past 10 years, there has been a global decline in the quality performance of primary healthcare buildings (Gambo and Musonda, 2021). For this analysis, only journal articles and review articles were selected. This is mainly because of the internationally accepted standard for publishing peer-reviewed research outputs (De Jong *et al*., 2015). The study also considered only publications written in English.

The selected articles were downloaded from Scopus and exported to AtlasTi. AtlasTi is a qualitative data analysis software commonly used to manage and analyse qualitative research data. It allows researchers to organise, code, and analyse various types of qualitative data, including text, images, audio, and video (Woods *et al*., 2015). AtlasTi was used to scope the literature to see the gaps, opportunities, and gaps for AI in healthcare asset information management.

Bibliometric analysis

Bibliometric analysis, widely used in different fields, uses mathematical statistics to quantitatively analyse and evaluate research trends between journals, authors, countries, institutions, and subject categories. This plays a significant role in researchers' quick grasp of literature development and critical information (He *et al*., 2020). This study used keyword co-occurrence, bibliographic coupling, co-citation, country, journal co-citation, co-occurrence, and clustering analysis as indications for the analysis. The co-occurrence of keywords study examines the conceptual framework of the field using the keywords provided by the author (He *et al*., 2017). Clustering analysis, keyword co-occurrence, and author co-citation analysis give a general overview of the subject field. Numerous science mapping methods are available for examining and displaying structural, dynamic, and temporal patterns and trends in the scientific literature of a knowledge domain. Each tool has unique advantages and capabilities regarding the kind of analysis it can conduct. VOSviewer was chosen as a result to analyse any domain thoroughly. The fundamental capability needed for creating, viewing, and navigating bibliometric networks is provided by the software programme VOSviewer (Darko *et al*., 2019).

Results

Annual publication and citation analysis

Figure 2 shows the yearly trend of publications and citations to show the research progress in Scopus-indexed articles related to the asset information modelling of healthcare-built infrastructure using AI.

Disciplines

Table 1 reviews the research disciplines and subject areas from which the various publications come, as derived from Scopus. Some of the publications are cross-disciplinary, indicating the integration of knowledge and methodologies from different subject areas, contributing to a holistic understanding of the research topic.

Keyword co-occurrence and cluster analysis

Topics were detected using keyword co-occurrence analysis using VOSviewer. A total of 1227 keywords were found. With a minimum occurrence of eight co-occurrences of keywords, 31 keywords occurred in clusters. Initially, there were 104 keywords. However, after a thesaurus analysis, there were 21. A VOSviewer thesaurus file can disregard terms, merge terms, synonyms, and abbreviated terms with complete terms, rectify spelling errors, and indicate that different names pertain to the same researcher. The study also performed data cleansing using a thesaurus file. Table 2 shows the most active keywords in the 104 analysed documents. The most co-occurring keywords from the database were healthcare (71), human (55), health facilities (22), organisation and management (18), and health service (15).

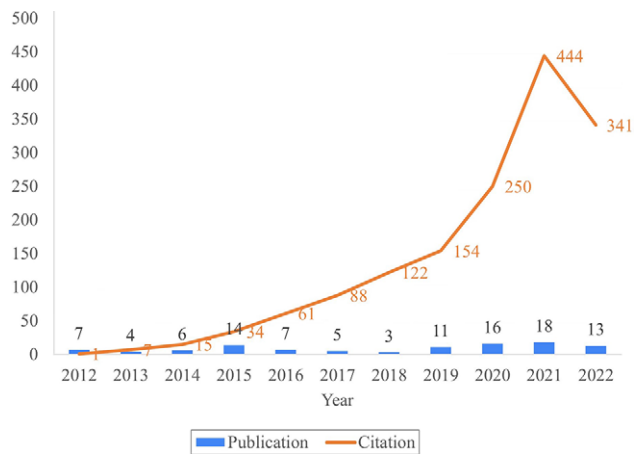

Figure 2. Citations per publication plotted against publication year

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Table 1. A review of research disciplines and subject area

Subject area	Count
Medicine	61
Engineering	30
Social sciences	17
Business, management, and accounting	16
Computer science	14
Chemical engineering	5
Biochemistry, genetics, and molecular biology	4
Decision science	
Material science	
Multidisciplinary	
Economics, econometrics, and finance	3
Environmental science	
Pharmacology, toxicology, and pharmaceutics	
Physics and astronomy	
Chemistry	2
Earth and planetary sciences	
Mathematics	
Nursing	
Psychology	
Arts and humanities	1
Immunology and microbiology	

Cluster 1: essential healthcare services

This cluster has eight keywords, with the average publication year for this cluster being around 2016–2018. This cluster points out that research on these terms has been ongoing in recent years, focusing on understanding disparities in healthcare access, socioeconomic influences, and controlled studies. According to Akter *et al*. (2021), the essential qualities of a resilient community are required, with a focus on infrastructure resilience. This cluster highlights how resource management might increase a community's capacity to withstand pandemics such as COVID-19. Emergency risk governance positively impacts the community's capacity to transform against vulnerabilities by influencing resource vulnerability and pandemic-time adaption processes. Community resilience also presupposes an inclusive and participatory planning process focusing on stakeholder identification and relationships (community involvement), risk assessment (physical and social capital/assets), goal setting, and ranking adaptation/mitigation methods.

The healthcare infrastructure must be adaptable to meet functional and spatial demands. This includes modularity and adaptability, which are both examples of flexibility. They facilitate adjustments to technological and epidemiological changes; as a result, it is an essential factor to consider while designing a hospital. Healthcare facilities should take certain practical measures to minimise the spread of infectious diseases by modifying their physical environments (Marmo *et al*., 2022). Healthcare facilities can enhance efficiency by demonstrating the information types that must be managed therein. While AI has the potential to minimise the labour hours required to manually find the necessary information, as noted by Demirdöğen *et al*. (2021), it is imperative to acknowledge that these advancements are significantly impeded by the

Table 2. Cluster analysis of the most used keywords in the 104 articles

Cluster ID	Keyword			
	Word	Occurrence	Total link strength	Average publication year
1				
3	Controlled study	10	63	2018
17	Healthcare disparities	11	70	2016
20	Household	8	57	2017
22	Major clinical study	10	75	2017
26	Pregnancy	9	65	2017
27	Procedures	9	52	2016
29	Socioeconomic factors	8	64	2016
30	Statistics and numerical data	11	76	2015
2				
4	COVID-19	11	33	2021
5	Delivery of healthcare	9	39	2018
15	Health service	15	85	2017
16	Healthcare	71	235	2019
19	Hospitals	13	24	2019
21	Human	55	230	2017
25	Pandemic	8	26	2021
28	Risk assessment	10	42	2018
3				
6	Economics	11	78	2016
8	Financial management	11	62	2016
10	Health care cost	11	72	2017
14	Health facilities	22	94	2016
24	Organisation and management	18	78	2015

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prevalent challenge of securing reliable and comprehensive data in this context.

Cluster 2: socioeconomic health infrastructure inequalities

This cluster consists of eight keywords, with an average publication of 2021. This indicates a recent focus on these topics, which aligns with the impacts on healthcare systems. The facility's state directly affects the end users (Salem and Elwakil, 2020). In disaster events, capacity planning and reconfiguring hospitals and healthcare facilities are needed (Aghapour *et al*., 2019). There is no agreed-upon definition of a resilient healthcare facility due to the growing need for natural disaster–resistant healthcare infrastructures. Making the wrong choice could result in the loss of life owing to a shortage of facilities or human resources because disaster healthcare resource planning is complex.

Khalid *et al*. (2022) noted disparities in health outcomes that might result from information platforms not designed with minority groups' specific health needs in mind. Research has also highlighted the potential of information platforms to improve health literacy by reducing information poverty. The disparities between the richest and poorest are exacerbated regarding access to and use of numerous preventative and curative health care. Ambitious goals have been set to empower and advance social, economic, and political justice in international strategies and achieve and sustain income development among the world's poorest populations (Akseer *et al*., 2016).

Cluster 3: healthcare infrastructure financing

This cluster has five keywords, with the average publication year for this cluster being 2016. It indicates a consistent focus on healthcare services, economics, financial management, and risk assessment in the past few years. Since healthcare facilities are typically operated with the financial backing of public organisations, an increase in quality and cost-effectiveness is needed. This encompasses workspace planning and management activities to ensure business continuity in a facility, accounting for 60% of the total life-cycle costs of a building (Demirdöğen *et al*., 2021). Wanigarathna *et al*. (2019) noted that AI could facilitate a range of data about the physical state of built assets, resources available for BAM, and the built asset's contribution within an organisation; AI could facilitate better-informed decision making about BAM.

A study was done by Roper *et al*. (2015) on the significant benefit and cost attributes of a radiofrequency identification (RFID) system for asset tracking in healthcare facilities. The literature review found some benefits, including improved staff productivity, quicker patient turnover, higher quality of care, and more cost savings. Real-time location system technologies allow hospitals to be prepared for emergencies requiring the immediate use of medical devices without delay and allow staff to determine the status, condition, and location of essential equipment, decreasing patient wait time.

People are paying more and more attention to the effective and intelligent management of existing buildings because of rapid economic development. An integrated solution was needed for healthcare projects that included complex operations to organise and document any changes made throughout the operation and maintenance phase. Wen *et al*. (2020) noted that deploying AI connectors to manage multiple data sources, information, and activities will ultimately help promote the organisation's business performance.

Discussions

Despite a significant study on healthcare facilities, no previous initiative has undertaken the task of collectively and quantitatively summarising these findings. Therefore, this study created an extensive fishbone diagram using a detailed scientometric review and in-depth content analysis. The subsequent section delves into the vital areas of research, identifies current research gaps, and suggests potential directions for future work in the context of leveraging AI in healthcare facilities' AIM.

Status quo of research

At the beginning of the 10-year analysis, various studies explored healthcare infrastructure planning (Mahadkar *et al*., 2012), stakeholder consultation (Terndrup *et al*., 2012), real-time locating systems (RTLS) (Kamel Boulos and Berry, 2012), and leveraged buyouts (Kim and McCue, 2012). These studies used qualitative methods such as interviews and literature reviews and quantitative methods such as longitudinal panel data analysis. Common findings included the importance of stakeholder consultation, the benefits of innovative technologies such as RTLS and data analytics, challenges in managing healthcare real estate, and the impact of financial factors on infrastructure planning.

By the end of the 10 years, the main research themes observed through the publications encompassed healthcare system and infrastructure, technology and innovation, stakeholder perspectives and engagement, the impact of COVID-19, and specific contextual studies. These papers explored various aspects of healthcare, including infrastructure planning, stakeholder consultation, innovative technologies, such as telemedicine apps and blockchain-enabled systems, and the effects of the COVID-19 pandemic on healthcare systems. In addition, they highlighted the importance of stakeholder engagement, considering diverse perspectives, and addressing context-specific challenges in improving healthcare delivery and management. Some papers focused on specific contexts or regions, such as rural African healthcare settings, the Afghan humanitarian crisis, Indian healthcare management systems, and Malaysian public healthcare. These studies aimed to address context-specific challenges and provide insights into improving healthcare practices and policies in those settings.

Moreover, 2022 healthcare research papers revealed trends in methodologies and topics – qualitative research methods, followed by quantitative approaches, focused on subjective experiences and stakeholder perspectives. Case studies and literature reviews were

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commonly used to examine real-world scenarios and synthesise existing knowledge. However, most papers focused on developed countries, highlighting the need for inclusivity in diverse healthcare contexts.

Trend analysis

An analysis of Scopus publications reveals several noteworthy trends in keyword usage. Among the identified keywords, 'asset' appears with a relatively lower frequency, occurring only four times since 2018. In contrast, 'information' was actively researched and appeared 24 times in 2020, while 'management' is also a popular subject, with 32 occurrences in 2018. The terms 'healthcare' and 'facility' have shown increasing significance, appearing 49 times in 2019 and 21 times in 2018. However, these words were not used independently but combined with other words. The analysis also highlights the emergence of two fields: 'infrastructure' and 'artificial intelligence'. These keywords have gained attention more recently, as they were found ten and nine times, respectively, in 2022. This shows that researchers have recently started investigating the applications of AI in managing asset information within healthcare based on the Scopus analysis.

When examining the overall keyword distribution, the most frequently occurring keyword is 'human', which appears 59 times with a total link strength weight of 383. On average, these publications have a publication year of 2017. Other frequently occurring keywords include 'humans' (49 occurrences) and 'health care delivery' (31 occurrences), with an average publication year of 2017. Regarding recent topics, 'COVID-19' stands out with the most recent average publication year of 2021, suggesting a significant focus on this subject due to the ongoing pandemic. Similarly, 'pandemic' has an average publication year of 2021, indicating its relevance in the analysed dataset. Thus, 'COVID-19' and 'pandemic' represent the more recent topics explored in these publications, while 'human' and 'humans' are the most frequently occurring keywords overall. Table 2 shows the co-occurence of words and those used in the search strings.

Figure 3 below shows how the keywords from the search string compare with those found in VOSviewer. Although words such as healthcare, management, information, facility, and asset appear in other words, the scatterplot considered the words singularly. Figure 3 shows the scatterplot of keywords distribution over the years.

Research methodologies

The studies provided valuable insights into aspects of healthcare, AI, and AIM. However, it is crucial to recognise and address their limitations to ensure a balanced interpretation of their findings. Familiarising oneself with the data gathering and analysis techniques employed can assist researchers in acquiring valuable understanding. For the data extraction, studies were categorised as design science research, mixed methods, qualitative, quantitative, and mixed methods. Qualitative research was the most popular methodology, and it was used in 57 studies. Within this category, the most common designs were literature reviews (11), followed

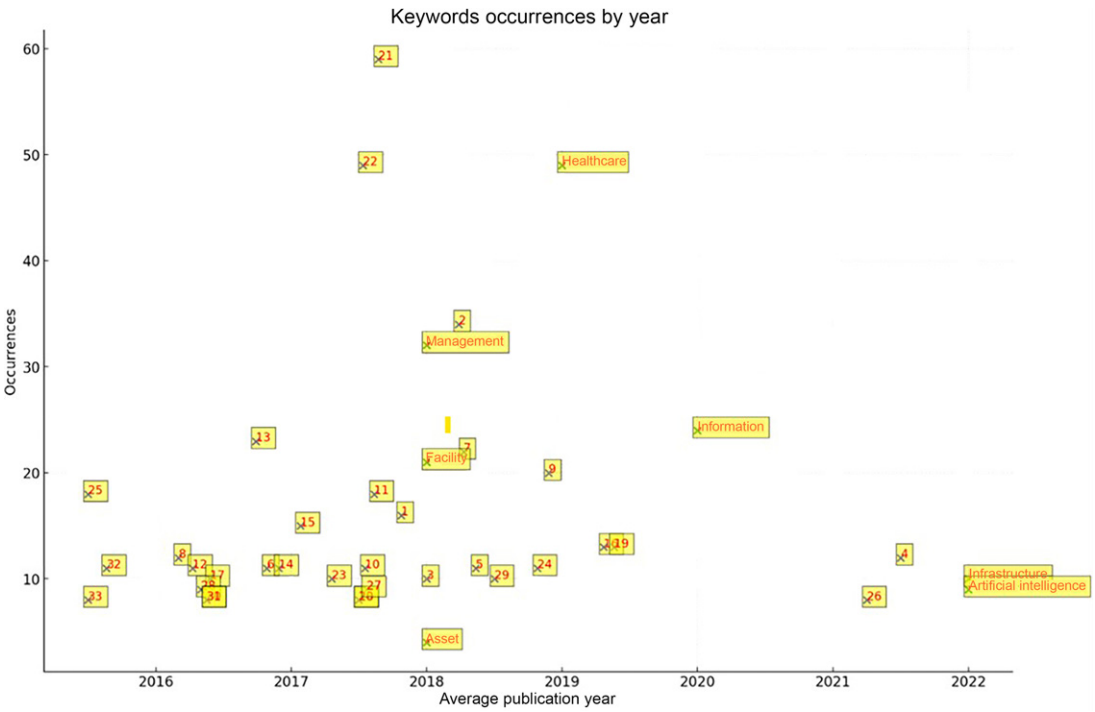

Figure 3. Scatter plots of the keyword's distribution by average year of mention

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by case studies (nine), literature reviews combined with expert interviews (five), and SLRs (five). Other qualitative designs included semi-structured interviews, retrospective analyses, and prospective studies, capturing detailed descriptions and perspectives from stakeholders involved in healthcare AIM.

Quantitative methods were employed in 31 studies, focusing on collecting and analysing numerical data to identify patterns, test hypotheses, and make predictions. The most common design, used in ten studies, involved collecting structured responses from participants through questionnaires, often analysed using principal component analysis. Simulation studies utilised in four studies created mathematical models to replicate real-world processes for predictive analysis and optimisation. Cross-sectional studies, mathematical models, and regression models–simulated scenarios and predicted outcomes each were used in three studies. Observational and retrospective studies were used in two studies, while another two employed longitudinal panel data analysis to track changes over time. Other designs included cost–benefit analysis, risk assessment, comparative analysis, bi-objective optimisation models, hybrid genetic algorithm–based fuzzy Markovian models, and stochastic frontier analysis, each used in one study.

Mixed methods were employed in 14 studies, combining qualitative and quantitative approaches. The most popular mixed methods are quantitative surveys, qualitative interviews, and cross-sectional designs. Studies also combined surveys with interviews and literature reviews. Another study combined literature reviews, case studies, and questionnaires. Policy analysis and needs assessment were each employed in some studies. Figure 4 shows the funnel measuring the methods used in the 104 articles.

One notable limitation common across several studies is the small sample size, which may limit the generalisability of the results. In the study of Miiro *et al*. (2022), only six key informant interviews and three focus group discussions were conducted, potentially failing to capture the full range of perspectives and experiences related to the topic. Limited samples such as these can introduce sampling bias and reduce the statistical power of the analyses, potentially affecting the findings' reliability.

In addition, it is important to consider the methodological limitations present in some studies. Essar *et al*. (2022) relied solely on a single policy analysis, influencing the researcher's perspectives and assumptions. This influence could impact the interpretation of challenges and opportunities identified in the study (Abrahams *et al*., 2022; Islam *et al*., 2022; McColl, 2014; Yang *et al*., 2020). The absence of a diverse range of stakeholders in this study might limit the comprehensiveness of the analysis. The reliance on self-reported data in quantitative research is also observed in these studies.

Particular attention should be given to the comparison between studies conducted in developing countries and those conducted in developed countries (Churi *et al*., 2022; Imran *et al*., 2020; Miiro *et al*., 2022; Valerio-Shewmaker *et al*., 2022). The varying development, infrastructure, and resource levels between these two contexts can significantly impact the findings and their applicability. It is crucial to recognise that the challenges and realities faced by healthcare systems in developing countries may differ significantly from those in developed countries. Therefore, caution should be exercised when extrapolating findings from one setting to another, as contextual differences may lead to different outcomes and conclusions.

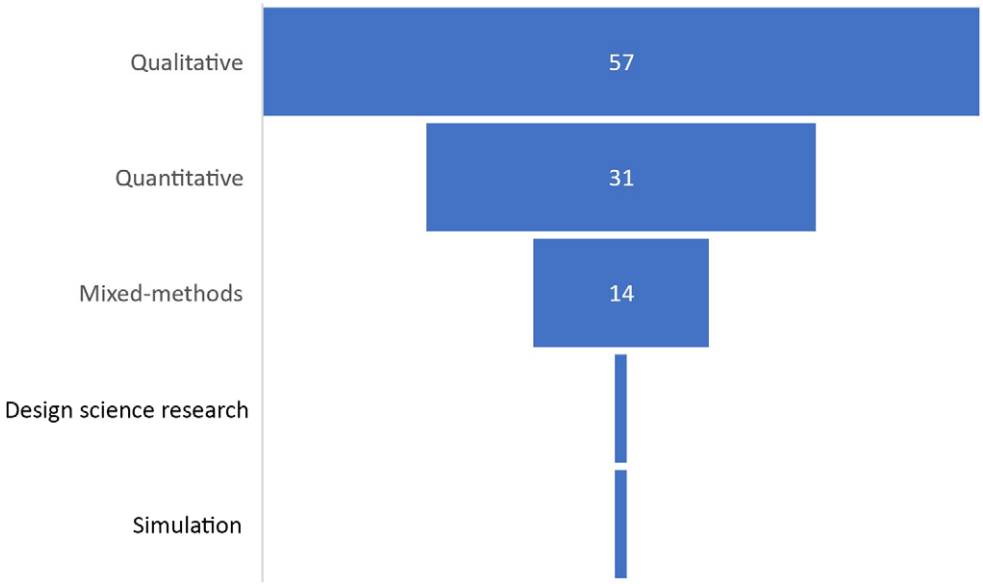

Figure 4. Research output measurement by methodological approach

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Only two of the 104 studies reviewed focused on healthcare facilities as their population and setting. These studies examined AI's application in healthcare, specifically in maintenance prioritisation and deterioration modelling (Demirdöğen *et al*., 2021). The objective was to enhance decision-making processes and optimise resource allocation. However, neither study provided information on the sample size nor measured the human element in its investigations.

Challenges and opportunities

Data quality and privacy

AI in healthcare institutions raises concerns about data privacy due to the sensitive nature of healthcare data. Strong data privacy safeguards such as encryption, access restrictions, and secure data storage procedures are necessary to protect this valuable information and prevent unauthorised access or exploitation.

Gonzalez-Granadillo *et al*. (2021) presented the Automated Cyber and Privacy Risk Management Toolkit (AMBIENT) as a valuable tool. AMBIENT provides automatic services tailored to user needs using sensors and computers to gather and analyse user information. Churi *et al*. (2022) investigated privacy concerns in India's healthcare knowledge management systems. However, the study had limitations, including focusing on a specific country and excluding patients and other stakeholders in the analysis, relying solely on healthcare professionals' perspectives. The study also lacked an in-depth exploration of healthcare professionals' privacy concerns, such as data security, potential discrimination, and profiling risks associated with their health data.

Interoperability

Interoperability is a significant challenge in healthcare asset information management and AI implementation. It involves integrating and exchanging data between different systems, which can be complex and time-consuming. The lack of interoperability among various data systems leads to fragmented data, posing significant challenges for efficient asset management and strategic decision-making processes. This fragmentation results from isolated data silos that emerge when systems cannot communicate and share information seamlessly. Consequently, organisations may struggle to comprehensively view their assets and related data, impairing their ability to make well-informed, strategic decisions. This hinders the optimal utilisation of assets and potentially jeopardises the effectiveness of management strategies and the realisation of organisational objectives.

Overcoming these challenges is crucial for healthcare facilities to leverage AI and achieve effective asset information management. Kawamoto *et al*. (2021) noted the challenges of establishing interoperable asset management record innovations, emphasising the need for multidisciplinary collaboration. However, there are gaps in research regarding the impact of interoperability on cost-effectiveness, implementation challenges, best practices, and socioeconomic records.

Standard

Standards are key for implementing asset information management and AI in healthcare by offering a common framework for data interoperability and system integration (Karpenko *et al*., 2018). The current ISO standards for whole life cycle and asset management, such as ISO 19650, 55000, and 16739, exist in the BIM space. Miiro *et al*. (2022) note that standardised systems could enhance data quality and interoperability here despite human resource scarcity, infrastructure deficits, and cultural barriers. However, their universal adoption is hindered by rapid technological changes and disparities between healthcare organisations, causing compatibility issues and standardisation gaps (Mbunge *et al*., 2022; Wijen, 2014). Yang *et al*. (2020) further stress the role of technical, operational, and financial standards in fostering private sector participation in healthcare through public–private partnerships (PPPs) in China. Ensuring technology compatibility, alignment with public objectives, and financial viability mitigates risks and promotes high-quality services. Clear and consistently enforced standards increase private sector engagement in healthcare PPPs.

Capacity and competencies

The rapid advancement of technology and changing healthcare asset management practices necessitate a skilled workforce to ensure efficient care delivery. However, a skills gap exists due to these advancements outpacing the capabilities of facilities management professionals (David and McNutt, 2019; Whysall *et al*., 2019). This issue can be addressed through continuous professional development, training initiatives, and collaboration between educational institutions and healthcare organisations (Harborth and Kümpers, 2022). Roper *et al*. (2015) point to a significant skills gap in adopting RFID technology in hospitals, necessitating focused training on using, reading, and interpreting data from RFID technology. Similarly, Che-Ani and Ali (2019) emphasise the impact of training and development on implementing proactive maintenance programs in healthcare facilities. This includes standardised training programs for facilities management professionals, including a lack of curriculum or certification processes for proactive maintenance.

Scalability and infrastructure

The successful integration of AI in healthcare facilities is faced with significant challenges related to scalability and infrastructure. Adequate physical infrastructure, such as space, servers, data centres, networking equipment, power supply, cooling systems, and backup mechanisms, is crucial to supporting AI systems. Ageing infrastructure assets contribute to backlogs and require additional investment to meet new regulations and technical requirements (Mills *et al*., 2015). Ensuring the security and privacy of patient data necessitates measures such as secure server rooms and data encryption systems. Addressing these physical infrastructure challenges is essential for efficiently and securely utilising AI technologies in healthcare (Bublitz *et al*., 2019). Digitally competent supply chains have the potential to enhance resource allocation, coordination, and management during public

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health emergencies. However, conducting comprehensive evaluations is essential to assess their effectiveness, scalability, and impact on cost-effectiveness and response efficiency.

Data governance

Data governance is a crucial challenge in implementing AI in healthcare facilities, encompassing the management and control of data, including its collection, storage, access, and usage (Janssen *et al*., 2020). Clear data ownership and accountability are vital aspects of data governance that healthcare facilities must establish. Determining who owns the data is essential for defining rights and permissions related to data access and usage.

Enhanced decision making

Enhanced decision making in healthcare facilities involves leveraging AI to analyse large amounts of data, identify patterns, and generate valuable insights to support decision-making processes (Elgendy *et al*., 2021). By utilising AI, healthcare facilities can make more accurate, efficient, and informed data-driven decisions. This approach optimises resource allocation, improves operational efficiency, and enhances overall performance in facility management. Ahmed *et al*. (2020) pointed out that the availability and accuracy of data on healthcare facility deterioration may influence the reliability and applicability of the proposed model.

Asset tracking and monitoring

Asset tracking monitors and manages physical assets such as machinery and furniture within facilities. By incorporating AI into asset tracking and management, especially in healthcare settings, operations can be enhanced (Riikkinen *et al*., 2018). AI allows real-time monitoring and data collection on asset movement, usage, and surrounding conditions. The data aid in better management and decision making regarding asset procurement and maintenance (Motlagh *et al*., 2020; Tang *et al*., 2019). Furthermore, AI can predict maintenance needs by identifying irregularities, thereby reducing equipment downtime and extending their lifespan, leading to improved operations and efficient resource allocation in healthcare facilities (Aqueveque *et al*., 2021).

Proactive analysis and maintenance

Proactive analysis and maintenance are significant opportunities for AI in asset information management in healthcare facilities (Mannino *et al*., 2021). By leveraging AI, healthcare organisations can collect and analyse data from various sources to identify potential problems with assets before they fail. This proactive approach helps prevent costly downtime and improves safety. AI algorithms can detect anomalies in real-time data, enabling early identification of issues and prompt maintenance interventions. Moreover, AI can utilise historical data to predict equipment failures or performance degradation, facilitating proactive maintenance planning and optimised resource allocation (Gallego-Madrid *et al*., 2022). Implementing AI-powered systems in healthcare facilities enhances operational efficiency, extends asset lifespan, and ensures the seamless delivery of high-quality care.

Research gaps

This section provides a comprehensive overview of the research gaps identified in AI for effective healthcare asset information management. These gaps have been identified through an extensive review of relevant literature from various sources. To clearly understand these research gaps, Table 3 outlines the criteria for identifying them, details about each gap, and the sources from which they have been derived. The criteria include the description of the research gap, its affected stakeholders, its scope, its manifestations, and its significance in the context of asset information management in healthcare facilities.

Implications

Implications for researchers

The research implications of AI for adequate healthcare facilities' AIM call for researchers to focus on several key aspects. First, researchers must dedicate their efforts to developing and refining AI-powered tools and applications specifically designed for healthcare facilities' AIM. This involves exploring and optimising automated asset inventory, maintenance schedules, and procurement systems. Second, researchers should contribute to developing programs that equip healthcare professionals with the necessary skills to utilise and manage AI tools and systems effectively. In addition, researchers are responsible for addressing ethical concerns related to AI in healthcare facilities' AIM, such as algorithmic bias, privacy protection, and data security. By conducting rigorous research in these areas, researchers can drive AI's advancement and responsible use in healthcare facilities' AIM.

Implications for practice

Healthcare institutions should consider implementing AI in asset management, which could improve efficiency and resource allocation. Successful implementation requires staff training, good data management practices, and regulatory compliance. AIM tool should consider its support capacity and experience in healthcare. A change management plan is necessary for a smooth transition, and post-implementation, the performance of the AI system should be continuously evaluated and improved as necessary.

Future research directions

Future research should emphasise several directions to bridge the gap between research focus and practical implementation. There is a need for research focused on developing AI-driven solutions tailored to healthcare facilities' unique needs and challenges. This includes addressing issues related to system interoperability, data integration, workflow optimisation, and user acceptance. Research efforts should evaluate the impact of AI on operational efficiency, cost-effectiveness, patient outcomes, and overall healthcare facility performance. This will provide valuable insights and evidence to support the adoption and implementation of AI in AIM. Moreover, research should explore emerging trends and technologies in AI that have the potential to enhance asset information management in healthcare facilities further.

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Table 3. Gap analysis of research needs in AI in healthcare AIM

Details	Identified gap	Sources
AI for effective healthcare asset information management	The definition of AI in the context of healthcare facilities' AIM. There is no definition among studies, causing confusion and inconsistency in comparing studies and evaluating their impact on healthcare AIM.	(Ahmed *et al.*, 2020; Atek *et al.*, 2022; Cai *et al.*, 2022; Islam *et al.*, 2022; Kawamoto *et al.*, 2021)
Researchers, healthcare facilities professionals, and built environment professionals of the assets	Theoretical exploration without sufficient research on practical applications of AI in healthcare AIM. There has been an insignificant amount of theoretical exploration without adequate research on the practical applications of AI in healthcare AIM.	(Che-Ani and Ali, 2019; Lai *et al.*, 2022; Salem and Elwakil, 2020; Wanigarathna *et al.*, 2019)
Healthcare facilities	Lack of comprehensive understanding regarding the cost factor of integrating AI into AIM.	(Beitelmal *et al.*, 2017; Cai *et al.*, 2022; Dainiak *et al.*, 2013; Li *et al.*, 2020; Younis *et al.*, 2013)
Implementation and analysis of AI systems in AIM	Ethical considerations such as bias, privacy, and security are unexplored. Need for research on specific applications and cost-effectiveness.	(Gonzalez-Granadillo *et al.*, 2021; Islam *et al.*, 2022; Miiro *et al.*, 2022; Mills *et al.*, 2015; Roper *et al.*, 2015; Schønning *et al.*, 2021)
To improve asset information management in healthcare facilities	A gap between research focus and practical implementation. Need for research on AI applications, considering interoperability, data integration, workflow optimisation, and user acceptance. There is a need for research that considers factors such as interoperability, data integration, workflow optimisation, and user acceptance.	(Alam, 2021; Gonzalez-Granadillo *et al.*, 2021; Kamel Boulos and Berry, 2012; Wanigarathna *et al.*, 2019)
Over the past ten years (2012–2022)	Past studies have not focused on AI for healthcare asset information management. Noteworthy trends in keyword usage highlight the emergence of AI and infrastructure but with less emphasis on 'asset'. More recent topics are the pandemic and COVID-19, which could influence AI applications in AIM.	(Essar *et al.*, 2022; Kamel Boulos and Berry, 2012; Kim and McCue, 2012; Mahadkar *et al.*, 2012; Miiro *et al.*, 2022; Terndrup *et al.*, 2012)

Asset information management (AIM) revolves around five pillars: people, processes, technology, data, and information. People oversee operations and data quality; processes ensure consistent data handling; technology aids in data collection and analysis; and static and dynamic data are visualised through reports and dashboards. Future research should explore AIM's integration within organisations, the impact of AI, human influences, and industry-specific practices.

Decision making

The implications of AI in healthcare facilities' AIM profoundly impact decision-making processes within healthcare facilities. AI-powered tools and applications provide valuable insights and support data-driven decision making in asset management. Researchers should focus on developing AI models and algorithms that improve decision-making capabilities in asset allocation, maintenance prioritisation, and resource optimisation. By harnessing the power of AI, healthcare professionals can make more informed and effective decisions, leading to improved asset utilisation and overall operational efficiency.

Collaboration and innovation

Effective collaboration and innovation are critical for advancing the application of AI in healthcare AIM. Researchers, healthcare professionals, and industry stakeholders should collaborate in interdisciplinary efforts to address challenges and drive innovation in the field. Collaboration can foster the exchange of knowledge, expertise, and resources, facilitating the development of practical and impactful AI solutions. Furthermore, fostering a culture of innovation encourages the exploration of novel ideas, approaches, and technologies in asset information management. This includes exploring the potential of emerging technologies such as ML, natural language processing, and robotics in optimising AIM processes. Researchers should focus on developing AI tools, addressing ethical concerns, and fostering collaboration and innovation to drive the effective application of AI in healthcare facilities' asset information management. By doing so, they can contribute to improved decision-making processes, enhanced collaboration, and, ultimately, better patient care outcomes.

Conclusions

AI applications in healthcare facilities' AIM hold immense potential for improving operational efficiency and effectiveness in the healthcare industry. This study aimed to conduct an exploratory bibliometric and systematic review to determine AI's status, trends, themes, challenges, and opportunities in healthcare facilities' AIM. A total of 104 publications from 2012 to 2020 were sourced from Scopus using predefined inclusion and exclusion criteria. To answer RQ1 on the status quo of AI for effective healthcare AIM, the study looked at the disciplines, the 10-year analysis of publications, the popular topics at the beginning and the end of the decade, and the

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methodologies used. To address RQ2 on trends and themes prevalent in AI for AIM, co-occurrences and analysis were performed to highlight categories of keywords used in the literature regarding the current topic. Q3 looked at opportunities and challenges associated with implementing AI in AIM within healthcare, ranging from data quality to proactive analysis and maintenance. Finally, insights derived from the analysis were used to answer RQ4 by outlining research gaps deserving further attention and research implications.

The findings suggest using AI to enhance asset management practices in healthcare aspects, such as optimising inventory, maintenance schedules, and procurement processes. The study also highlighted gaps in understanding the cost-effectiveness of implementing AI in healthcare AIM and the ethical considerations associated with AI, including bias, privacy, and security. Further research is needed to evaluate the economic benefits and return on investment of AI in AIM and to develop robust ethical guidelines for responsible AI implementation. A gap was identified between research focus and practical implementation, emphasising the need for dedicated research on AI applications that consider interoperability, data integration, workflow optimisation, and user acceptance. The study revealed that previous research has not extensively focused on AI for healthcare asset information management, with emerging trends emphasising AI and infrastructure but with less emphasis on the 'asset' aspect.

By addressing these research gaps and implications, researchers, policymakers, and healthcare professionals can advance the field of AI in healthcare facilities' AIM. Implementing AI-powered solutions tailored to the unique needs of healthcare facilities, considering ethical considerations, and evaluating their cost-effectiveness will result in improved asset management practices, enhanced decision-making processes, and, ultimately, more efficient and effective healthcare operations.

The findings of this review serve as a valuable resource for informing future research directions. They guide researchers in addressing the identified gaps and further advancing the field of AI in healthcare facilities' asset information management. In addition, decision makers can benefit from this review as it offers insights into the current research and trends. These insights can aid in making informed decisions when implementing AI technologies to optimise asset information management practices in healthcare.

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